

MA313 : Linear Algebra I
Week 7: Dimension and Rank

Dr Niall Madden

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These slides were produced by Niall Madden, based on ones by [Tobias Rossmann](#).



Outline

1 1: Coordinates (again)

- A graphical interpretation

2 2: Isomorphisms

- Linear Transformations
- Invertible matrices
- Coordinate mappings for $\mathbb{R}^n$

3 3: Dimension

- The definition

4 4: Spaces with same dimension

- Dim of subspaces
- The Basis Theorem

5 5: Rank and Nullity

6 Exercises

For more details, see

- ▶ Chapter 7 (Linear Independence) of Linear Algebra for Data Science:
<https://shainarace.github.io/LinearAlgebra/linind.html>
- ▶ Sections 4.5 (Dimension of a Vector Space) of Lay:
<https://ebookcentral.proquest.com/lib/nuig/reader.action?docID=5174425>

Assignment 3

There was a technical issue with WeBWork over the weekend, only resolved yesterday evening. So I've extended the deadline by 48 hours, to 5pm, Wednesday 19 Oct 2022.

Communication Skills : Progress Report

Your progress report is due 5pm, Friday 21 Oct.
Information on the content and structure are on Blackboard.

New assignment.... see Blackboard.

1: Coordinates (again)

MA313 Week 7: Dimension and Rank

Start of ...

PART 1: Coordinates

This is continued from the end of last week's classes

1: Coordinates (again)

Last week we learned that, if the sequence $\mathcal{B} = (b_1, b_2, \dots, b_n)$ be a basis of V , then we can write any vector $x \in V$ as a unique linear combination of the vectors in $\mathcal{B}$. That is:

- For any x , there is a set of real numbers $c_1, c_2, \dots, c_n$, such that

$$x = c_1 b_1 + c_2 b_2 + \dots + c_n b_n.$$

- Furthermore, there is only one set of numbers $c_1, c_2, \dots, c_n$ for which this is true.

1: Coordinates (again)

Since this collection of numbers is so important, it has a name: the **coordinate vector** of $x \in V$ relative to $\mathcal{B}$ is

$$[x]_{\mathcal{B}} := \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix},$$

where $c_1, \dots, c_n \in \mathbb{R}$ is the unique sequence with

$$x = c_1 b_1 + \cdots + c_n b_n$$

The function $V \rightarrow \mathbb{R}^n$, $x \mapsto [x]_{\mathcal{B}}$ is the **coordinate mapping** determined by $\mathcal{B}$.

1: Coordinates (again)

Example

Let

$$\mathcal{B} = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right)$$

be the standard basis of $\mathbb{R}^n$.

Then $[x]_{\mathcal{B}} = x$ for all $x \in \mathbb{R}^n$.

Hence, taking coordinate vectors *generalises* extracting the components of a vector in $\mathbb{R}^n$.

1: Coordinates (again)

Example

Let $B = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$.

1. B is a basis of $\mathbb{R}^2$.

First, note $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are linearly independent : $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq c \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ for any c .

Second: show $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ spans $\mathbb{R}^2$. That is, if $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ we can solve for c_1 & c_2 . Here $x_1 = c_1 + c_2$ & $x_2 = 2c_2$.
So $c_2 = \frac{x_2}{2}$ and then $c_1 = x_1 - \frac{x_2}{2}$.

1: Coordinates (again)

Example

Let $\mathcal{B} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$.

2. Write down the coordinate mapping determined by $\mathcal{B}$. It is a linear transformation, so also write down the matrix of the linear transformation.

We have that

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{\mathcal{B}} = \begin{bmatrix} x_1 - x_2/2 \\ x_2/2 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Coordinate
mapping

$\uparrow$
matrix of
the linear trans.

1: Coordinates (again)

Example

Let $\mathcal{B} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$.

2. Write down the coordinate mapping determined by $\mathcal{B}$. It is a linear transformation, so also write down the matrix of the linear transformation.

Note: $\det(A) = \frac{1}{2} \neq 0$

so A^{-1} exists.

Also: $\begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 - 2(\frac{1}{2}) \\ 0 + (\frac{1}{2})(2) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Standard
basis

1: Coordinates (again)

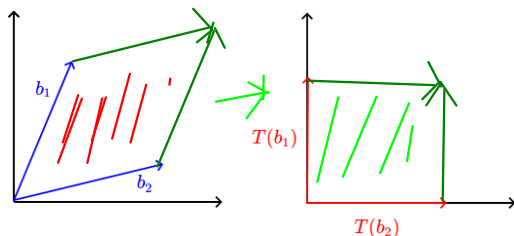
A graphical interpretation

Suppose that $\mathcal{B} = (b_1, b_2)$ is a basis of $\mathbb{R}^2$.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2, x \mapsto [x]_{\mathcal{B}}$ be the associated coordinate mapping.

Then $T(b_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $T(b_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Note that $\mathcal{B}$ defines a parallelogram. The coordinate mapping T “stretches”, “rotates”, and perhaps “reflects” it into a square!



2: Isomorphisms

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PART 2: Isomorphisms

2: Isomorphisms

INVERTIBLE FUNCTIONS

Let X and Y be sets and let $f: X \rightarrow Y$ be a function.

Then the following are equivalent:

- ▶ f is **invertible**, i.e., there exists $f^{-1}: Y \rightarrow X$ such that $f^{-1}(f(x)) = x$ for all $x \in X$ and $f(f^{-1}(y)) = y$ for all $y \in Y$.
- ▶ f is **one-to-one** and **onto**. (Also called “**injective**” and “**surjective**”).

Moreover, if f is invertible, then the function f^{-1} is **uniquely** determined.

2: Isomorphisms

Definition (ISOMORPHISM)

An **isomorphism** from a vector space V to a vector space W is an invertible linear transformation $V \rightarrow W$.

We say that V and W are **isomorphic** if there exists an isomorphism between them.

2: Isomorphisms

Example

- For any vector space V , the **identity map**

$$\text{id}_V: V \rightarrow V, \quad x \mapsto x$$

is an isomorphism.

Hence, every vector space is isomorphic to itself.

- Given any basis $\mathcal{B} = (b_1, \dots, b_n)$ of V , the coordinate mapping

$$V \rightarrow \mathbb{R}^n, \quad x \mapsto [x]_{\mathcal{B}}$$

is an isomorphism.

(We saw in Part 3 that this is an invertible linear transformation).

Theorem

Let U , V , and W be vector spaces.

Let $S: U \rightarrow V$ and $T: V \rightarrow W$ be linear transformations.

Then:

- ▶ *$T \circ S: U \rightarrow W, x \mapsto T(S(x))$ is a linear transformation.*
- ▶ *If S and T are isomorphisms, then so is $T \circ S$.*

That is: if U is isomorphic to V and V is isomorphic to W , then U is isomorphic to W .

Theorem

If $T: V \rightarrow W$ is an isomorphism of vector spaces, then so is $T^{-1}: W \rightarrow V$.

Hence, if V is isomorphic to W , then W is isomorphic to V .

Question

Can we relate this to matrices and vectors?

Let A be an $n \times n$ matrix.

Then the function

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad x \mapsto Ax$$

is a linear transformation.

It is invertible if and only if A is an invertible matrix. In that case, T^{-1} is the function

$$\mathbb{R}^n \rightarrow \mathbb{R}^n, \quad y \mapsto A^{-1}y.$$

Summary

- ▶ $m \times n$ matrices correspond to linear transformations $\mathbb{R}^n \rightarrow \mathbb{R}^m$.
- ▶ An $n \times n$ matrix is invertible if and only if the corresponding linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$ is an isomorphism. In that case, the inverse of the linear transformation corresponds to the inverse matrix.

Question...

Can there be an isomorphism $\mathbb{R}^n \rightarrow \mathbb{R}^m$ when $m \neq n$?

Let $\mathcal{B} = (b_1, \dots, b_n)$ be a basis of $\mathbb{R}^n$.

Then

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad x \mapsto [x]_{\mathcal{B}}$$

and its inverse $T^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ are both linear transformations from $\mathbb{R}^n$ to itself.

Question...

What are the matrices corresponding to T and T^{-1} ?

By definition:

$$T(x) = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \iff x = c_1 b_1 + \cdots + c_n b_n \iff T^{-1} \left(\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right) = x.$$

Hence, for $i = 1, \dots, n$,

$$T^{-1} \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right) = b_i$$

so the matrix of T^{-1} is $A := [b_1 \cdots b_n]$, and the matrix of T is therefore A^{-1} .

3: Dimension

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PART 3: Dimension

3: Dimension

In many parts of mathematics, the concept of “dimension” can be very difficult and subtle... and even counter-intuitive.

Example (From analysis)

There exists a continuous function from the unit interval $[0, 1]$ onto the unit square $[0, 1]^2$. (In fact, there exists a continuous surjection $[0, 1] \rightarrow [0, 1]^n$ for each n .)

Such functions are called **space-filling curves**

Fortunately, such issues don't arise in linear algebra...

3: Dimension

What should “dimension” mean/imply?

1. The **dimension**, $\dim V$ is a number associate with to each vector space V .
2. If V and W are isomorphic, then we want that $\dim V = \dim W$.
3. We want that $\dim \mathbb{R}^n = n$ for each n .
4. In fact, if $(b_1, \dots, b_n)$ is a basis of V , then we want that $\dim V = n$.

So, would it make sense to take that as a definition, at least when V is finitely generated? It would require that any two bases of V contain the same number of vectors...

3: Dimension

Theorem

Let $(b_1, \dots, b_n)$ be a basis of V . Then every sequence consisting of at least $n + 1$ vectors in V is linearly dependent.

Clearly, if the set of vectors is

$\{b_1, b_2, \dots, b_n, d\}$, we can write d as

a linear combination of $b_1, \dots, b_n$. So set is linearly dependent.

Otherwise, if the set is

$\{u_1, u_2, \dots, u_n, u_{n+1}\}$. Write each

$u_1, \dots, u_n$ in terms of $b_1, \dots, b_n$.

3: Dimension

Theorem

Let $(b_1, \dots, b_n)$ be a basis of V . Then every sequence consisting of at least $n + 1$ vectors in V is linearly dependent.

For a full proof, see Theorem 9 in Chap 4 of the text-book by Lay et al.

3: Dimension

Corollary

If V has *some* basis consisting of precisely n vectors, then *every* basis of V consists of precisely n vectors.

This is an immediate consequence of the previous theorem.

Example: $\mathbb{R}^3$ has a basis $\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$.

- Any other basis for $\mathbb{R}^3$ has 3 vectors too.
- Eg $\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right)$ could not be a basis, since it has too few vectors

Definition (DIMENSION)

The **dimension** of V is, a vector space, is

$$\dim V = \begin{cases} 0, & \text{if } V = \{0\}, \\ n, & \text{if } V \text{ has a basis } (b_1, \dots, b_n), \\ \infty, & \text{if } V \text{ is not finitely generated.} \end{cases}$$

Example

- ▶ $\dim \mathbb{R}^n = n$: the standard basis consists of n vectors.
- ▶ $\dim \mathbb{P}_n = n + 1$: the sequence $(1, t, t^2, \dots, t^n)$ is a basis.
- ▶ $\dim \mathbb{P} = \infty$ because this space is not finitely generated. (Why?)

Eg $\mathbb{P}_2$ has as a basis $(1, t, t^2)$.

So $\dim \mathbb{P}_2 = 3$.

4: Spaces with same dimension

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PART 4: Spaces with the same dimension

4: Spaces with same dimension

FACT

Isomorphic vector spaces have the same dimension.

Isomorphic means that, there is an invertible mapping between the spaces. Call this $T: V \rightarrow W$.
So, if $(b_1, b_2, \dots, b_n)$ is a basis for V , then $(T(b_1), T(b_2), \dots, T(b_n))$ is a basis for W .

P.T.O

4: Spaces with same dimension

FACT

Isomorphic vector spaces have the same dimension.

In more detail: suppose $\dim(V) = n$,
and $\dim(W) = m$ and V & W are isomorphic.
We want to show $m = n$.

Since $\dim(V) = n$, denote a basis for it
as $(b_1, b_2, \dots, b_n)$. Call the isomorphic mapping
 T . Since $\{b_1, b_2, \dots, b_n\}$ are linearly independent
in V , so $\{T(b_1), T(b_2), \dots, T(b_n)\}$ are linearly
independent in W . So $m \geq n$.

P.T.O.

4: Spaces with same dimension

FACT

Isomorphic vector spaces have the same dimension.

Similarly, If $(w_1, w_2, \dots, w_m)$ is a basis for W , then $\{T(w_1), T(w_2), \dots, T(w_m)\}$ are linearly indep. in V . So $n \geq m$

But if $m \geq n$ and $n \geq m$,
it must be that $m = n$.

Question

How are the concepts “subspace” and “dimension” related?

Example

The subspaces of $\mathbb{R}^3$, sorted by dimension, are:

- ▶ 0-dimensional: just $\{0\}$.
- ▶ 1-dimensional: subspaces spanned by a single non-zero vector. That is, such subspaces are lines through the origin.
- ▶ 2-dimensional: planes passing through the origin.
- ▶ 3-dimensional: just $\mathbb{R}^3$.

Theorem

Let V be a finitely generated vector space. Let H be a **subspace** of V .
Then:

- ▶ H is also finitely generated.
- ▶ $\dim H \leq \dim V$.
- ▶ Any linearly independent sequence of vectors in H can be extended to a basis of V .
- ▶ $\dim H = \dim V$ if and only if $H = V$.

Theorem (The Basis Theorem)

Let $n = \dim V$ satisfy $1 \leq n < \infty$. Let $v_1, \dots, v_n \in V$. Then following are equivalent:

1. $(v_1, \dots, v_n)$ is a basis of V .
2. $v_1, \dots, v_n$ are linearly independent.
3. $V = \text{span} \{v_1, \dots, v_n\}$.

So, if we have a space of known dimension, n , and n vectors $(v_1, \dots, v_n)$ we can check if that is a basis by checking any one of ② or ③.

Theorem (The Basis Theorem)

Let $n = \dim V$ satisfy $1 \leq n < \infty$. Let $v_1, \dots, v_n \in V$. Then following are equivalent:

1. $(v_1, \dots, v_n)$ is a basis of V .
2. $v_1, \dots, v_n$ are linearly independent.
3. $V = \text{span} \{v_1, \dots, v_n\}$.

Eg $\left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right)$

Not a basis.

5: Rank and Nullity

MA313 Week 7: Dimension and Rank

Start of ...

PART 5: Rank and Nullity

5: Rank and Nullity

Definition (RANK and NULLITY)

Let A be an $m \times n$ matrix.

- ▶ The **rank** of A is the dimension of its column space:
 $\text{rank } A := \dim \text{Col } A.$
- ▶ The **nullity** of A is the dimension of its null space:
 $\text{nullity } A := \dim \text{Nul } A.$

When we were finding bases for the column space and null space of a matrix, we found that, if a matrix has p pivot columns then

$$\text{rank } A = p$$

and

$$\text{nullity } A = n - p.$$

5: Rank and Nullity

Theorem

Rank-Nullity Theorem $\text{rank } A + \text{nullity } A = n$.

In particular, $\text{rank } A$ and $\text{nullity } A$ determine one another.

This is one of the most famous and important results of linear algebra!

Example

Confirm that $\text{rank } A + \text{nullity } A = n$ where

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

5: Rank and Nullity

The reduced row echelon form of A is

$$A' = \begin{bmatrix} 1 & 2 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

↑ ↑ ↑
Pivot cols

A has
rank
3.

Since cols 1, 3 & 4 of A' are linearly indep in A' , so too col 1, 3 & 4 of A are linearly indep. So they are a basis for A 's column space. That is

$\text{Col } A = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Also $\dim \text{Col } A = 3$.

5: Rank and Nullity (This slide added after class)

Recall, if $x \in \text{Nul } A$ (ie $Ax=0$) then

$x \in \text{Nul } A^T$ (ie $A^T x=0$). Thus

$$\begin{bmatrix} 1 & 2 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$x_1 + 2x_2 - 3x_5 = 0$$

$$x_3 + x_5 = 0$$

$$x_4 = 0$$

non pivot cols:
associated variables
are "free".

Solving the system we get

- x_5 is free

- $x_4 = 0$

- $x_3 = -x_5$

- x_2 is free

- $x_1 = -2x_2 + 3x_5$

5: Rank and Nullity

So, any vector of the form $\begin{bmatrix} -2x_2 + 3x_5 \\ x_2 \\ -x_5 \\ 0 \\ x_5 \end{bmatrix}$ is

in the null space of A' (and A).

$$\text{So } \text{null}(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

$$\text{So } \dim(\text{null}(A)) = 2.$$

5: Rank and Nullity

In summary, this calculation suggests

- * $\dim \text{Col } A = (\text{number of pivot columns of } A')$
- * $\dim \text{Nul } A = (\text{number of non-pivot columns of } A')$
- * so $\dim \text{Col } A + \dim \text{Nul } A = (\text{number of columns of } A' = n).$

5: Rank and Nullity

There is a version of this for vector spaces.

Theorem

Rank-Nullity Theorem (abstract form) Let $T: V \rightarrow W$ be a linear transformation between vector spaces. Then

$$\dim \operatorname{Ker} T + \dim \operatorname{Ran} T = \dim V.$$

(We'll return later to have a closer look at the required translation between matrices and linear transformations).

Finished here, but some details added after class.

5: Rank and Nullity

Returning the matrices...

Theorem (Invertible Matrix Theorem)

Let A be an $n \times n$ matrix.

Then the following are equivalent:

1. *A is invertible, i.e. there exists an $n \times n$ matrix B such that $AB = I_n = BA$.*
2. $\text{rank } A = n$.
3. $\text{nullity } A = 0$.
4. *The columns of A form a basis of $\mathbb{R}^n$.*

Exercises

Q1. Let $\mathcal{B} = \left(\begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \end{bmatrix} \right)$. Show that $\mathcal{B}$ is a basis of $\mathbb{R}^2$ and find the vector $x \in \mathbb{R}^2$ with coordinate vector $[x]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$.

Q2. Let $\mathcal{B} = \left(\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} \right)$. Show that $\mathcal{B}$ is a basis of $\mathbb{R}^3$ and find the vector $x \in \mathbb{R}^3$ with coordinate vector $[x]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$.

Q3. Find the dimension of this subspace of $\mathbb{R}^3$:

$$\left\{ \begin{bmatrix} s - 2t \\ s + t \\ 3t \end{bmatrix} : s, t \in \mathbb{R} \right\}$$

Exercises

Q4. Find the dimension of this subspace of $\mathbb{R}^4$.

$$\left\{ \begin{bmatrix} 2c \\ a - b \\ b - 3c \\ a + 2b \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$$

Q5. Find the dimension of the subspace of $\mathbb{R}^2$ spanned by

$$\begin{bmatrix} 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 15 \end{bmatrix}.$$

Q6. Find the dimensions of $\text{Nul } A$ and $\text{Col } A$, where

$$A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Exercises

Q7. Find the rank of these matrices:

$$\begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 3 & 4 & -1 & 2 \\ 2 & 6 & 6 & 0 & -3 \\ 3 & 9 & 3 & 6 & -3 \\ 3 & 9 & 0 & 9 & 0 \end{bmatrix}.$$

- Q8. If the null space of a 4×6 matrix A is 3-dimensional, what is the dimension of the column space of A ?
- Q9. If the null space of an 8×7 matrix A is 5-dimensional, what is the dimension of the column space of A ?